# Harish G

**Forward Deployed Engineer | AI Workflows in Production, Agent Orchestration, Platform Reliability**

Bengaluru, India | +91 7892855850 | harishgreddy.work@gmail.com | LinkedIn | GitHub | harishg.com

## Profile

I am the engineer who stays after the demo works. At CortexCraft.ai I built the voice-agent platform our other shipped products now drive through its REST API, and most of my hours there went into routing, Gemini socket reconnects and a TLS failure traced to NAT hairpinning rather than into new features. I work in Python and TypeScript, I sit with whoever owns the problem until the real bottleneck is named, and I treat model output as a claim to verify rather than a result to display.

## Core Skills

| | |
|---|---|
| **Languages** | Python · TypeScript / JavaScript · Java · Kotlin · SQL |
| **AI Systems** | LLM Integration · Agents & Tool Calling · RAG · Output Guardrails · Prompt Frameworks · Agent Orchestration · Gemini Live · Eval Fixtures |
| **Backend** | FastAPI · Node.js / Express · Spring Boot · Celery · WebSockets · REST API Design |
| **Frontend** | React · Vite · Tailwind CSS · Streamlit |
| **Production** | Request-ID Tracing · JSON Logs · Prometheus Metrics · pytest in CI · Docker & Compose · Jenkins · GitLab CI · Runbooks & Rollback |
| **Data** | PostgreSQL · MongoDB · Redis · Neo4j · Supabase / pgvector |

## Experience

**Software Developer, CortexCraft.ai** Jan 2026 – Present

- Built and maintain the voice-agent platform (589 of 626 commits on the SaaS platform, 67 of 75 on the core engine): Gemini Live phone agents over Twilio and Plivo, per-agent RAG grounding, prompt generation, transcripts, and an external REST API that other products call instead of forking the stack, so one fix to the call pipeline reaches every product at once.
- Own production reliability of that pipeline. Fixed outbound routing, added reconnect handling for dropped Gemini 1006 sockets, tightened goodbye detection so calls end cleanly, and traced a Docker TLS failure to NAT hairpinning rather than restarting past it.
- Shipped LeadCall AI into client deployments on the platform: multi-tenant outbound calling, Celery dial and callback workers behind Redis dial locks, pitch documents re-indexed for RAG so the script matches what the client sells, per-tenant credentials encrypted at rest, and an append-only audit log of every operator action.
- Design the guardrails and tool-calling orchestration that keep model output constrained. Tools are declared functions, extracted call data is coerced to its declared type before anything downstream reads it, and escalation to a human is an explicit path rather than a fallback.

**Freelance Web & App Developer (Self-Employed)** 2023 – 2026

- Delivered projects end to end for local clients, from the first scoping conversation through launch and the fixes afterwards: React/Vite front ends, Python and Node.js services, automation scripts and deployment support.

## Selected Projects

**JiraFlow Agent and AI Code-Review Gate** *Python, FastAPI, Jira API, GitHub API, Ollama, React*

- Turns a Jira ticket into a branch, a code change and a pull request, one `ai/scrum-<id>` branch per ticket whose commit cites the issue. The agent proposes and a human merges, so a wrong edit costs a rejected PR rather than a broken main branch.
- The review gate is the part an agent cannot argue past: high-risk changes are blocked from approval outright, a rejection requires a written reason that is fed back into regeneration, every reviewer action lands in an append-only audit log, and nothing auto-merges.

**KisanVoice AI** *Python, FastAPI, React, MongoDB, WhatsApp Flows, Google STT/TTS, Prometheus, Jenkins*

- 172 of 192 commits. Migrated a live platform off its Node backend onto FastAPI and off Sarvam onto Google STT/TTS behind a config-driven registry of 24 locales (10 Indian), so adding one is a config row rather than a deploy, and no survey in flight was interrupted. Then operated it: request-ID tracing, JSON log lines, a Prometheus metrics endpoint, Jenkins CI and a runbook with a rollback path.

**AI Personal CFO and Talo** *Kotlin, Jetpack Compose, Hilt, Room/SQLCipher, ML Kit, Gemini Live*

- Sole author, 159 commits, 34 Gradle modules, 199 test files. Twelve deterministic engines compute every figure and the model is only permitted to explain them, on a written rule: numbers from math, words from AI.
- An output guardrail rewrites any number in model text that is absent from the tool result as `[unverified]`. An MCP-style in-process tool host exposes the engines through a JSON registry carrying `inputSchema`, `readOnlyHint` and `destructiveHint`; chat reads through the registry and never writes. The OCR and SMS parsers are scored against frozen evaluation fixtures, so a model regression shows up as a failing test.

## Education & Ways of Working

**B.E., Information Science and Engineering**, Vivekananda Institute of Technology, Bangalore

Feature branches and conventional commits; GitLab CI with 132 pytest files as the merge gate on the voice-agent platform. Ship a thin first version early, then harden it where the usage says to.